

STATE BOARD OF TECHNICAL EDUCATION AND TRAINING
TELANGANA

DIPLOMA EXAMINATION (C-24)

C-24-OCT-2025

SEMESTER III , MID-II EXAM

AI/CCB/CCP/CPS/CS

CS-303

Data structures through 'C'



**PCODE
243030**

Duration: 1 Hour

[Total Marks: 20]

004

PART-A

004

Instructions:

1. Answer the following questions.
2. Each question carries **ONE** mark.

4 X 1 = 4

1. What is the Best case time complexity of Binary Search?
2. List out any four sorting algorithms.
3. List the terminologies associated with stacks.
4. List the operations performed on a queue.

004

PART-B

004

Instructions:

1. Answer the following questions.
2. Each question carries **THREE** marks.

2 X 3 = 6

- 5(a). Explain briefly the procedure of bubble sort

---- OR ----

- 5(b). What is a Linear Search? Write the main steps involved in Linear Search?

- 6(a). List any three applications of stacks.

---- OR ----

- 6(b). Evaluate the following postfix expression in a neat tabulated form: $2\ 3\ 1\ * + 9$

PART-C

Instructions:

1. Answer the following questions.
2. Each question carries **FIVE** marks.

2 X 5 = 10

004

004

04 7(a). Write a C program to implement the selection sort.

----- OR -----

7(b). Write a C program to implement Binary Search (Assume the input array is in Sorted order)

8(a). Explain about sparse matrix.

----- OR -----

8(b). Write a C program to implement stacks using arrays.